

Ask any audiophile for recommendations on a good pair of bluetooth headphones and all you may get in response is a scoff and maybe a chuckle. While it is good enough for calls and does manage decent quality audio in the car (the credit goes to the amplifier), Bluetooth is a lossy transmission protocol. Bluetooth uses its own compression protocol and codecs to



ensure that the packet sizes fit the data bandwidth. The receiver pieces back the packets to play back the compressed audio. Bluetooth physically doesn't have enough bandwidth to transmit high-quality audio, and even if it did, it would consume more power. The key variable at play here is the codec, which is most

commonly SBC (low complexity SubBand Coding). Newer codecs like A2DP and AptX promise better (CD-quality) audio, using a method called time domain ADPCM (Adaptive Differential Pulse-Code Modulation). Basically, they enable smaller size samples, so that a higher sampling rate can fit in the same bandwidth, resulting in better quality audio.

Bluetooth v5 is the latest specification soon to be released that doubles the speed and increases the range up to four times. Although the upgrades are aimed towards IoT devices, it also means better audio quality for all!

## Kleer

Developed in 2006 specifically for wireless audio, Kleer is a proprietary wireless transmission protocol. It enables transmission of lossless audio, as it offers a maximum data bandwidth of 2.37Mbps. Compare this with the fact that CD-quality 16-bit 44KHz stereo audio requires only 1.42Mbps. Kleer also has other advantages like a lower radio bandwidth requirement and hence lower battery consumption, along with the ability to transmit to up to 4 different receivers simultaneously. Unfortunately for all its advantages, Kleer hasn't proliferated the market and support remains limited.



## To wrap things up...

You've already covered the latter leg of this auditory journey in the previous chapters (assuming you are a sequential reader). With that, our acoustic voyage is complete and you, dear reader, are privy to the secrets behind audio tech wizardry. Another step closer to becoming a knowledgeable audiophile! **d**